

USB Type-C Interface

The diagram illustrates the USB Type-C interface circuit. It shows the connection between a USB Type-C receptacle (J5) and a USB Type-C connector (U11) via a USB Type-C to USB-A adapter (U11).

USB Type-C Receptacle (J5):

- VBUS: A4
- CC1: B5
- CC2: B5
- D-: A7
- D+: A6
- D+: B6
- SHIELD: S1
- GND: A1

USB Type-C Connector (U11):

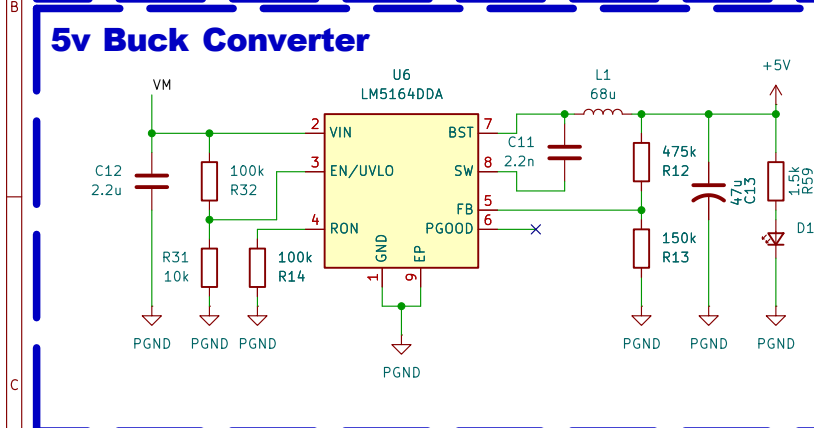
- 1: VBUS
- 2: D-
- 3: D+
- 4: GND
- 5: VBUS
- 6: D-

USB Type-C to USB-A Adapter (U11):

- VBUS: Vusb
- D-: USB_D-
- D+: USB_D+
- GND: GND

Other Components:

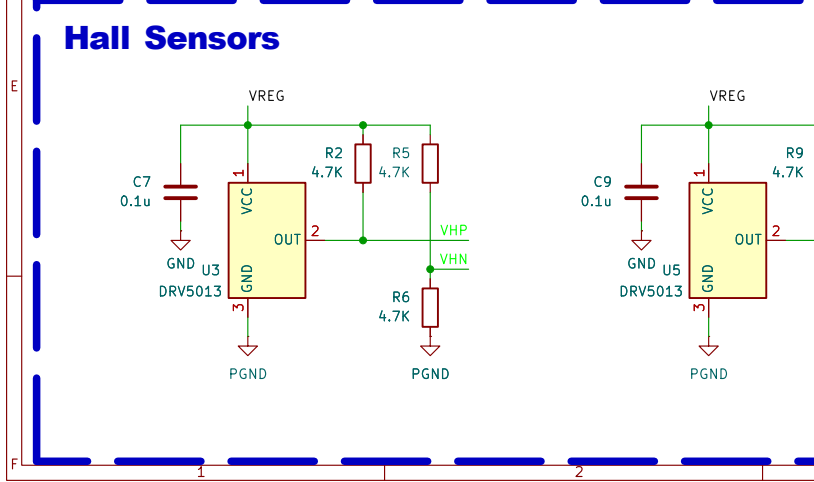
- R15: 5.1K
- R30: 5.1K
- C27834: 5.1K
- C27834: 5.1K
- TestPoint TP4
- USB_DP
- USB_DN



The diagram illustrates the electrical connections for the Boot/reset switches and the 24Mhz Oscillator.

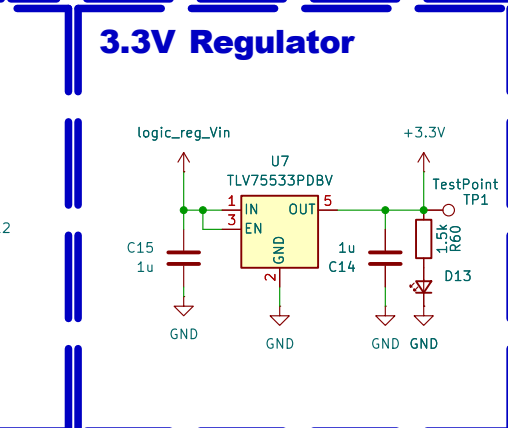
Boot/reset switches: This section shows two push-button switches, SW1A and SW2A, both labeled as SW_Push_Dual_x2. SW1A is connected to the NRST pin (green) and has a 10k resistor (R38) to +3.3V. SW2A is connected to the BOOT pin (green) and has a 10k resistor (R37) to +3.3V. Both switches have one terminal connected to GND.

24Mhz Oscillator: This section shows the oscillator circuit. It includes a crystal (Y1, labeled Crystal_GND23) connected between the OSC_IN and OSC_OUT pins. Two 15pF capacitors (C29 and C2) are connected from each oscillator pin to GND.



Power Input & Protection

The diagram illustrates the power input and protection circuit for a shielded component. The shield component, labeled 'SHIELD', has three pins: P (Power), N (Neutral), and SH1 (Shield). The P pin is connected to a red wire labeled '1' which passes through a red rectangular fuse labeled 'F1 Fuse' before reaching the '+BATT' terminal. The N pin is connected to a blue wire labeled '2' which goes to the cathode of a red diode labeled 'D1 1.5SMCxxCA'. The SH1 pin is connected to a green wire labeled 'SH1' which also goes to the cathode of the diode. The anode of the diode is connected to the '+BATT' terminal, and the cathode is connected to the 'PGND' terminal. This configuration provides over-voltage protection for the power input line.



SWD Debug Interface

The diagram illustrates the SWD Debug Interface. A J2 connector is shown with the following pins and connections:

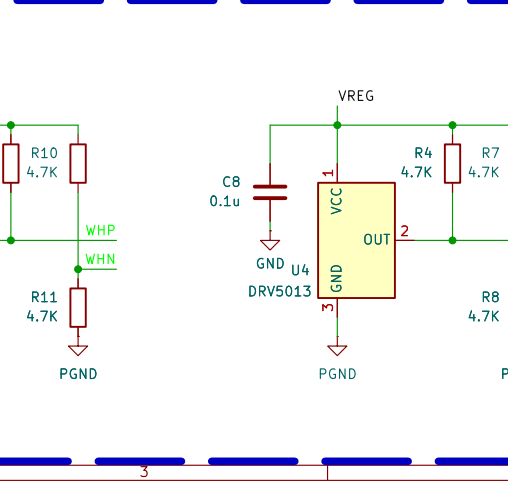
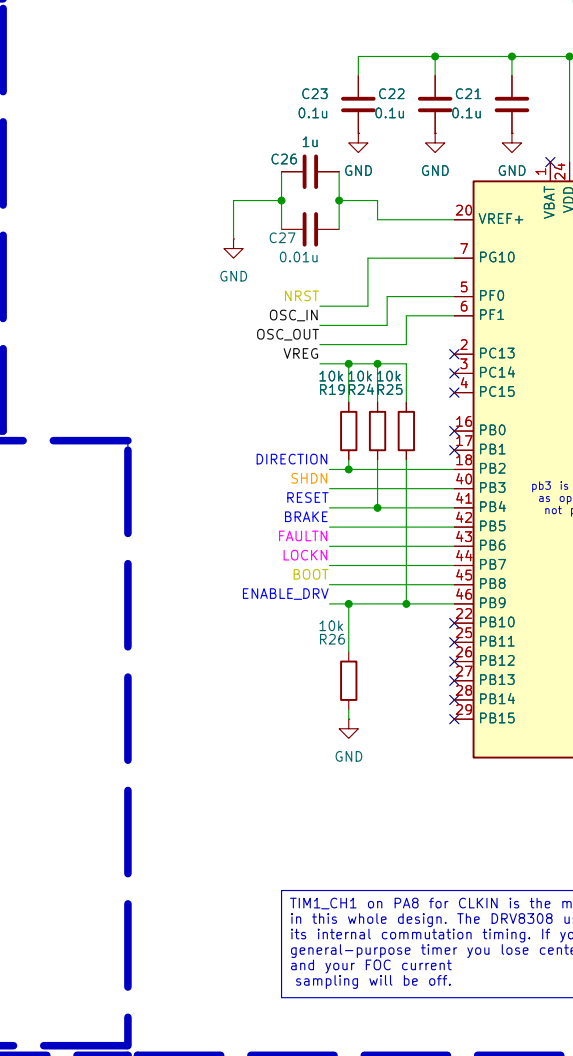
- Pin 1: VCC, connected to +3.3V.
- Pin 2: SWCLK, connected to pin 4 of the microcontroller header.
- Pin 3: RESET, connected to pin 3 of the microcontroller header.
- Pin 4: SWDIO, connected to pin 2 of the microcontroller header.
- Pin 5: GND, connected to GND.
- Pin 6: SWO, connected to pin 6 of the microcontroller header, which is marked with an 'X'.

The microcontroller header is labeled with the following pins:

- 3: NRST
- 4: SWCLK
- 2: SWDIO
- 6: SWO (marked with an 'X')

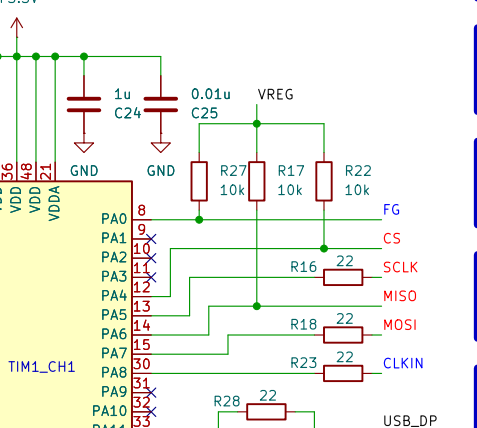
Additional labels in the diagram include:

- C_OUT (on the left edge)
- P 8 (on the left edge)
- Conn_ARM_SWD_TagConnect_TC2030-NL (below the connector)

[illegible]

Power orring

The diagram illustrates a power orring circuit. It features two diodes, D2 and D3, both labeled MBR5340. The anode of D2 is connected to a +5V input, and the anode of D3 is connected to a Vusb input. The cathodes of both diodes are connected to a common output node labeled logic_reg_Vin. This node is shown with a red arrow pointing upwards, indicating the output voltage. The entire circuit is enclosed in a blue dashed rectangular frame.

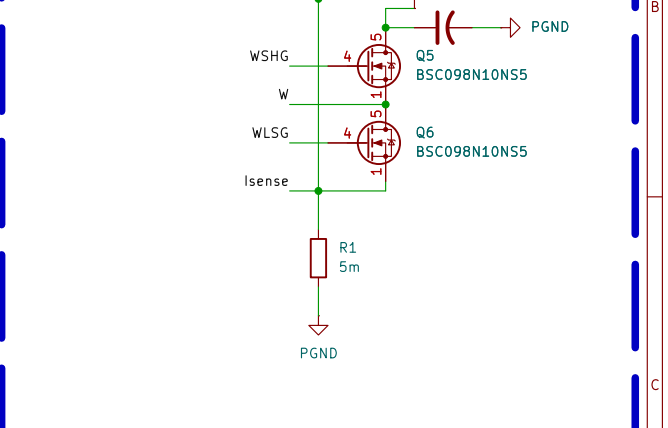


Power MOSFET Stage

The diagram illustrates a Power MOSFET Stage with four MOSFETs (Q1, Q2, Q3, Q4) and three capacitors (C16, C17, C18). The MOSFETs are arranged in a vertical stack, with their gates connected to input signals and their drains connected to a common output node (VM) and ground (PGND). The MOSFETs are labeled Q1, Q2, Q3, and Q4, all of type BSC098N10NS5. The capacitors are labeled C16, C17, and C18, all of type 220u. The input signals are USHG, U, ULSG, VSHG, V, and VLSG. The output node is labeled VM. The ground connection is labeled PGND.

Key components and connections:

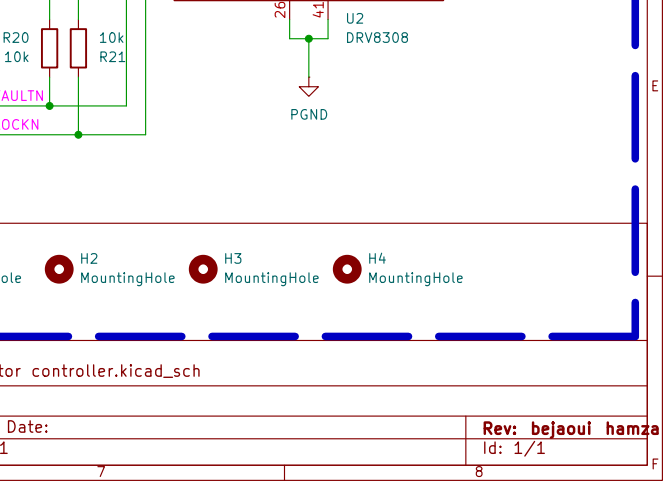
- MOSFETs:** Q1, Q2, Q3, Q4 (BSC098N10NS5)
- Capacitors:** C16, C17, C18 (220u)
- Input Signals:** USHG, U, ULSG, VSHG, V, VLSG
- Output Node:** VM
- Ground:** PGND



FOC unit

Pinout diagram for the FOC unit. The diagram shows the connections for various signals to the FOC unit pins. The pins are numbered 1 through 30. The connections are as follows:

- Pin 1:** CLK
- Pin 2:** S
- Pin 3:** I0SI
- Pin 4:** I1SO
- Pin 5:** ND
- Pin 6:** ESET
- Pin 7:** BRAKE
- Pin 8:** LKIN
- Pin 9:** IRECTION
- Pin 10:** NABLE_DRV
- Pin 11:** C10
- Pin 12:** ND
- Pin 13:** G
- Pin 14:** REG
- Pin 15:** VINT
- Pin 16:** VCP
- Pin 17:** VM
- Pin 18:** VREG
- Pin 19:** UHSG
- Pin 20:** U
- Pin 21:** ULSG
- Pin 22:** VSHG
- Pin 23:** V
- Pin 24:** VLSG
- Pin 25:** WSHG
- Pin 26:** W
- Pin 27:** WLSG
- Pin 28:** Isense
- Pin 29:** UHP
- Pin 30:** UHN
- Pin 31:** VHP
- Pin 32:** VHN
- Pin 33:** WHP
- Pin 34:** WHN
- Pin 35:** FGFB
- Pin 36:** FGINN_TACH
- Pin 37:** FGINP
- Pin 38:** FGOUT
- Pin 39:** FAULTn
- Pin 40:** LOCKn
- Pin 41:** GND
- Pin 42:** GND



Motor driver Supply Decoupling

The diagram illustrates the supply decoupling for a motor driver, divided into three main sections: Input, VM (Motor), and VREG (Regulator).

- Input Section:** Shows the input supply lines UHP (green) and UHN (red) connected to PGND. A red box indicates a component (likely a fuse or inductor) in series with the input.
- VM Section:** Shows the VM supply line connected to PGND. It includes a 470uF capacitor (C1) and a 0.1uF capacitor (C6) in parallel, with a TestPoint TP3 connected to the VM line.
- VREG Section:** Shows the VREG supply line connected to PGND. It includes a 1uF capacitor (C5) and a 0.1uF capacitor (C4) in parallel, with a TestPoint TP2 connected to the VREG line.
- FB1 Section:** Shows the FB1 (Ferrite Bead) connected between GND and PGND, with a red box indicating the bead component.